

Rajat Sharda

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TECHNICAL SKILLS:

Languages: Java, Python, C++, Javascript TypeScript, Redux, HTML, CSS, React, NodeJs, Tailwind CSS,

Tools/Frameworks: Github, Visual Studio, Postman, NeMo-Guardrails, OpenAI, Gemini, Claude, AWS, Azure, GCP

Databases: MYSQL, SQLite, Firebase, MongoDB, Sanity

EDUCATION:

B.Tech. - Delhi Technological University (CGPA: 8.50)

December 2021 - Current

Major - Electronics and Communications Engineering, **Minor** - Computer Science Engineering t

PROFESSIONAL EXPERIENCE:

Software Engineer | TechiZen India

July 2025 - Present

- Added a new service to create any number of questions related to a career context from one or multiple sample questions/description.
- Integrated creation of activities in Learnosity and addition of assessments on web from created questions, reducing manual effort

Software Engineering Intern | Amazon India

January 2025 - June 2025

- Architected and productionised the HLD and LLD for the migration of a Theft-Fraud-Abuse-Wastage detection rule improving data consistency & service downtime by 10%. Migrated 2 data syncing jobs from AWS Lambda functions to EventBridge
- Created and modified 3 APIs to aggregate expenses for over 1M+ employees. Leveraged AWS StepFunctions and EventBridge for workflow automation; and Lambda, OpenSearch, DynamoDB and S3 instead of legacy AWS Redshift, reducing \$80k per year.
- Enabled AWS Cloudwatch and Metrics, added alarms for new packages and deployed via AWS CDK, reducing manual effort

AI Engineering Intern | TechiZen India

May 2024 - December 2024

- Built and deployed robust AI solutions, including a RAG-based chatbot using Google Vertex AI, AWS Bedrock, LangChain, Ollama handling 1,500+ contextual queries/month and an AI-powered quiz generator that reduced content creation time by 83% through automated assessment generation from PDFs and YouTube and an image recognition feature utilizing ChatGPT-4o
 - Developed backend APIs and services for multi-modal LLM orchestration, metadata-aware retrieval utilizing AWS Lambda, API Gateway, GCP Discovery Engine, and Learnosity integration; delivered streamlined assessment validation and scalable data pipelines.
 - Designed SEO-optimized website leveraging Next.js, TypeScript, TailwindCSS, Redux. Integrated Microsoft Clarity, leading to a 65% boost in scroll depth and 40% higher CTR.
 - Successfully implemented the LLM for the client's AI teaching instructor, leveraging NeMo Guardrails to restrict the instructor to the topic and check for profanity, jailbreaking and hallucinations. This provided a success rate of over 90% for the LLM.
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ACADEMIC PROJECTS:

Vidya AI | *React, NodeJs, AWS, NeMo Guardrails, Langchain, Gemini-1.5-Flash* - [Link](#)

September 2024

- Modeled and built a smart AI teacher using Google's Gemini, capable of implementing Socratic Teaching method about any given topic, while ignoring all other profanity and jailbreaking attempts. Qualified for round 2 in Google's GenAI Hackathon
- Wrote custom actions and flows for enabling cross-questioning and follow up capabilities, along with knowledge-base provider action to work with user given topics/subjects. Aggregated a score of 95% success rate.

Portfolio | *React, TypeScript, Redux, Tailwind CSS, NeMo Guardrails, Groq API* - [Link](#)

May 2024 - June 2024

- Developed a personal portfolio website inspired by macOS, integrating features such as Shut Down, Sleep, Restart, draggable-resizable applications like NotePad, Safari, and VS Code. Utilized Redux for robust state management
- Engineered "Siri", an AI Assistant powered by Groq's Llama-3 capable of answering queries about me, while maintaining conversational memory, averaging 85% faster response time

Invictus Website (DTU Technical Fest'24) | *React, NextJs, Tailwind* - [Link](#)

December 2023 - January 2024

- Directed a team of six in designing and launching the official website for INVICTUS'24, the tech-fest of DTU; achieved a 60% increase in online registrations within the first week.
 - Enabled features for event registration, team formation, and participation summary for all fest activities, resulting in 20% increase in total footfall
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EXTRACURRICULAR ACTIVITIES:

- **Organizer & Events Lead** at VHIAAN'24, INVICTUS'24, and **Co-Head** at INVICTUS'23 (Tech-Fest of DTU)
- Freelanced as **SME** for Physics and Maths, responsible for cross-checking 10k+ high-school level questions
- Active member of Madhurina (**Music Society of DTU**) with a diploma in Violin (**2.8M+** views on **YouTube**)